COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	07/08//26	**DATE SUBMITTED:**	07/10/26
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ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm

Start
Declare variables: q1, q2, q3, total, average, passingScore
Set passingScore to 7.5
Read q1, q2, q3
Calculate total: q1 + q2 + q3l and average: total /3
If average >= passingGrade, display “Passed”, else display “Failed”

Pseudocode

```
BEGIN
        DECLARE q1, q2, q3, total, average as FLOAT
        DECLARE passingScore as FLOAT = 7.5

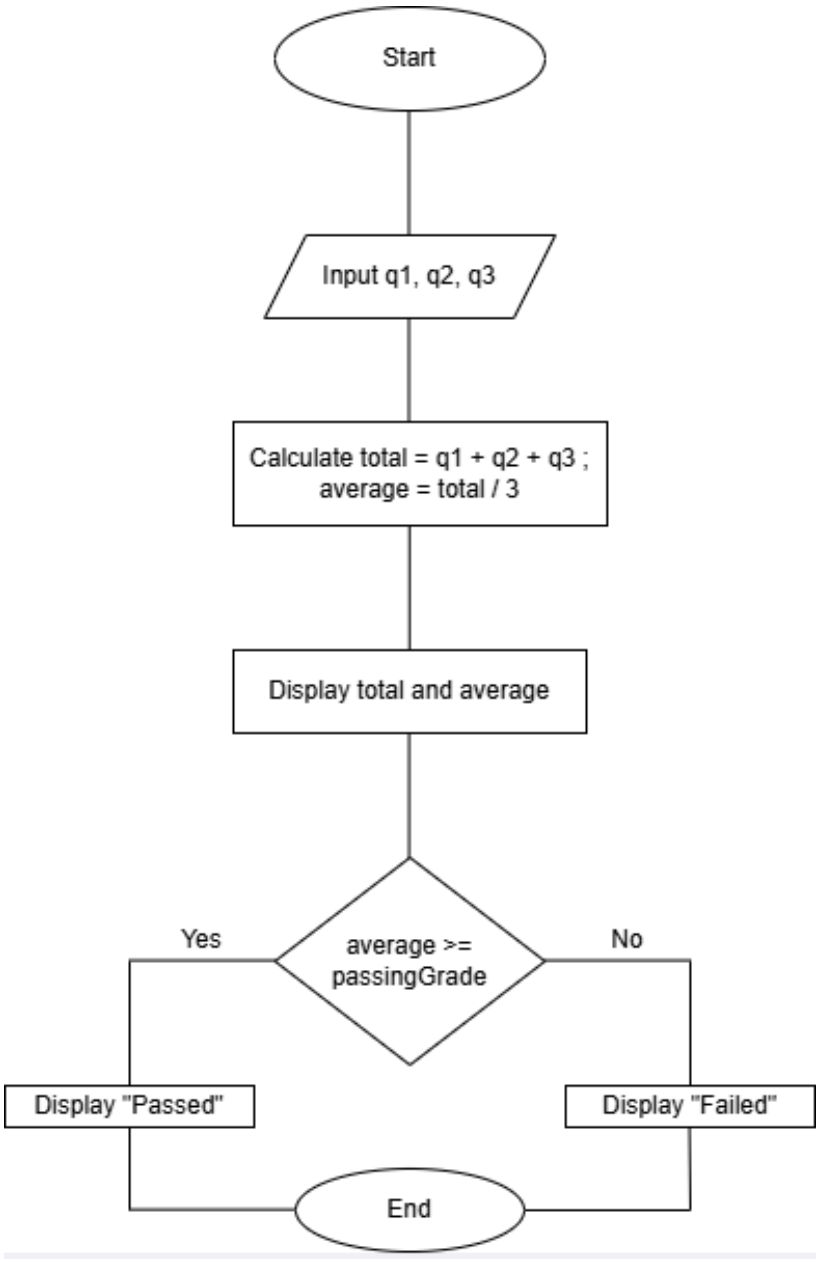
        OUTPUT “Quiz 1: “
        INPUT q1
        OUTPUT “Quiz 2: “
        INPUT q2
        OUTPUT “Quiz 3: “
        INPUT q3

        total = q1 + q2 + q3
        average = total / 3

        OUTPUT “Total: “, total
        OUTPUT “Average: “, average

        IF average >= passingGrade THEN
                OUTPUT “Passed”
        ELSE
                OUTPUT “Failed”
        END IF
END
```

Flowchart



Code Snippet and Output

C: > Users > Administrator > Desktop > CPE21S1 - Villaruel > quiz.cpp > mai

```cpp
//Prince Benjamin S. VIllaruel

#include <iostream>
#include <iomanip>
using namespace std;


int main() {
    float q1, q2, q3;
    float total, average;
    float passingGrade = 7.5;


    cout << "Quiz 1: ";
    cin >> q1;
    cout << "Quiz 2: ";
    cin >> q2;
    cout << "Quiz 3: ";
    cin >> q3;

    total = q1 + q2 + q3;
    average = total / 3;

    cout << fixed << setprecision(2);
    cout << "Total: " << total << endl;
    cout << "Average: " << average << endl;

    if (average >= passingGrade)
        cout << "Passed" << endl;
    else
        cout << "Failed" << endl;

    return 0;


}
```

```
PS C:\Users\Administrator\Desktop\CPE21S1 - Villaruel>
Quiz 1: 10
Quiz 2: 7.5
Quiz 3: 8
Total: 25.50
Average: 8.50
Passed
```

Code Analysis

For this first program, I managed to create a simple system that calculates the total and average of the three quizzes scores, then added a simple feature that checks if the average score meets the passing grade of 7.5 (estimating that the total score of each quiz is 10 points) , displaying the status as the result. Overall, this was a good recall of core C++ concepts and managed to challenge my capability by adding a simple feature in the program.

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm

Start
Declare item1, item2, item, price1, price2, price3, total, payment, change
Read item1, item2, item3, price1, price2, price3
Calculate total = price1 + price2 + price3
Display total
Read payment
If payment >= total: change (payment - total), display all the transactions made.
Otherwise, display insufficient payment and amount still owed.
End

Pseudocode

```
BEGIN
        DECLARE item1, item2, item3 AS STRING
        DECLARE price1, price2, price3, total, payment, change AS FLOAT

        OUTPUT “Enter name of Item 1: “
        INPUT item1
        OUTPUT “Enter price of : “, item1
        INPUT price1

        OUTPUT “Enter name of Item 2: “
        INPUT item2
        OUTPUT “Enter price of : “, item2
        INPUT price2

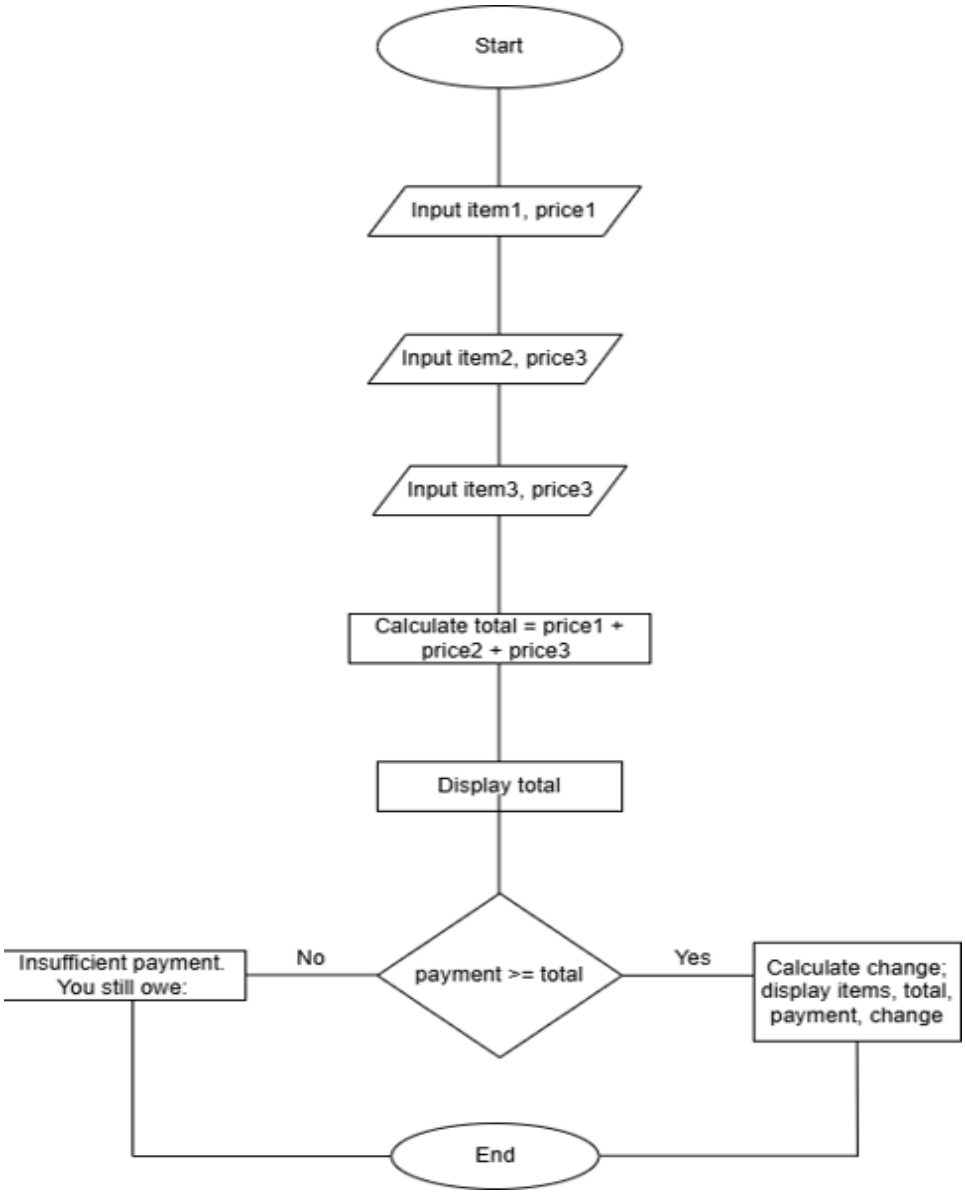
        OUTPUT “Enter name of Item 3: “
        INPUT item3
        OUTPUT “Enter price of : “, item3
        INPUT price3

        total = price1 + price2 + price3

        OUTPUT “ Total amount to pay: “, total
        OUTPUT “Enter payment amount: “
        INPUT payment
```

```
        IF payment  >= total THEN
                change = payment - total
                OUTPUT item 1, ":  ", price1
                OUTPUT item 2, ":  ", price2
                OUTPUT item 3, ":  ", price3
                OUTPUT "Total:  ", total
                OUTPUT "Payment: ", payment
                OUTPUT "Change: ", change
        ELSE
                OUTPUT "Insufficient payment.  You still owe:  ", (total - payment)
        END IF
END
```

Flowchart



Code Snippet and Output

```cpp
//Prince Benjamin S. Villaruel
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    string item1, item2, item3;
    float price1, price2, price3;
    float total, payment, change;

    cout << "Enter name of Item 1: ";
    cin >> item1;
    cout << "Enter price of " << item1 << ": ";
    cin >> price1;

    cout << "Enter name of Item 2: ";
    cin >> item2;
    cout << "Enter price of " << item2 << ": ";
    cin >> price2;

    cout << "Enter name of Item 3: ";
    cin >> item3;
    cout << "Enter price of " << item3 << ": ";
    cin >> price3;

    total = price1 + price2 + price3;

    cout << fixed << setprecision(2);
    cout << "Total amout to pay: " << total << endl;
    cout << "Enter payment amount: ";
    cin >> payment;

    if (payment >= total) {
        change = payment - total;

        cout << item1 << ": " << price1 << endl;
        cout << item2 << ": " << price2 << endl;
        cout << item3 << ": " << price3 << endl;
        cout << "Total: " << total << endl;
        cout << "Payment: " << payment << endl;
        cout << "Change: " << change << endl;
    } else {
        cout << "INsufficient payment. You still owe: "
            << (total - payment) << endl;

    }

```

```
Enter name of Item 2: Bread
Enter price of Bread: 25
Enter name of Item 3: Candy
Enter price of Candy: 10
Total amout to pay: 110.00
Enter payment amount: 150
Coke: 75.00
Bread: 25.00
Candy: 10.00
Total: 110.00
Payment: 150.00
Change: 40.00
```

Code Analysis

For the second activity, I created a basic cashiering system that takes the names and prices of the three products, computes the total, accepts a payment, and calculates the change while displaying the full transaction. I also challenged myself in this activity, by adding a feature that checks if the payment is sufficient before proceeding to the overall process. In the end, this was a good exercise to help me regain my knowledge in C++ and at the same boost my confidence in programming and solving complex problems.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used grammarly to check and correct the grammar of this activity.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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